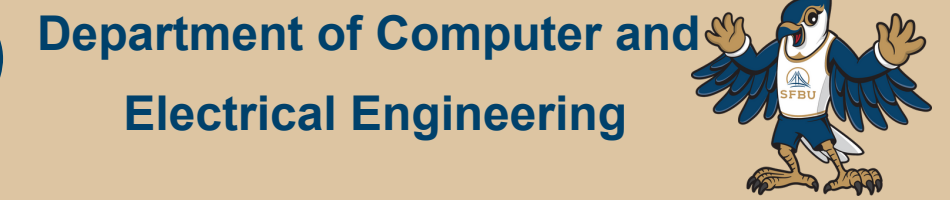




Capstone Project: TWOS (Travel WithOut Stress)



AI-based Travel Companion

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Project Advisor: Professor Ahmed Banafa

Why TWOS?

Today, planning a trip is surprisingly stressful. Travelers often have to jump between multiple websites and apps, compare endless options, and spend hours organizing the details. We started this project to solve that pain point.

TWOS (Travel WithOut Stress) takes care of the hard part for you.

By understanding the user's natural language, TWOS generates a complete, optimized travel plan saving time, reducing frustration, and simplifying the entire planning process.

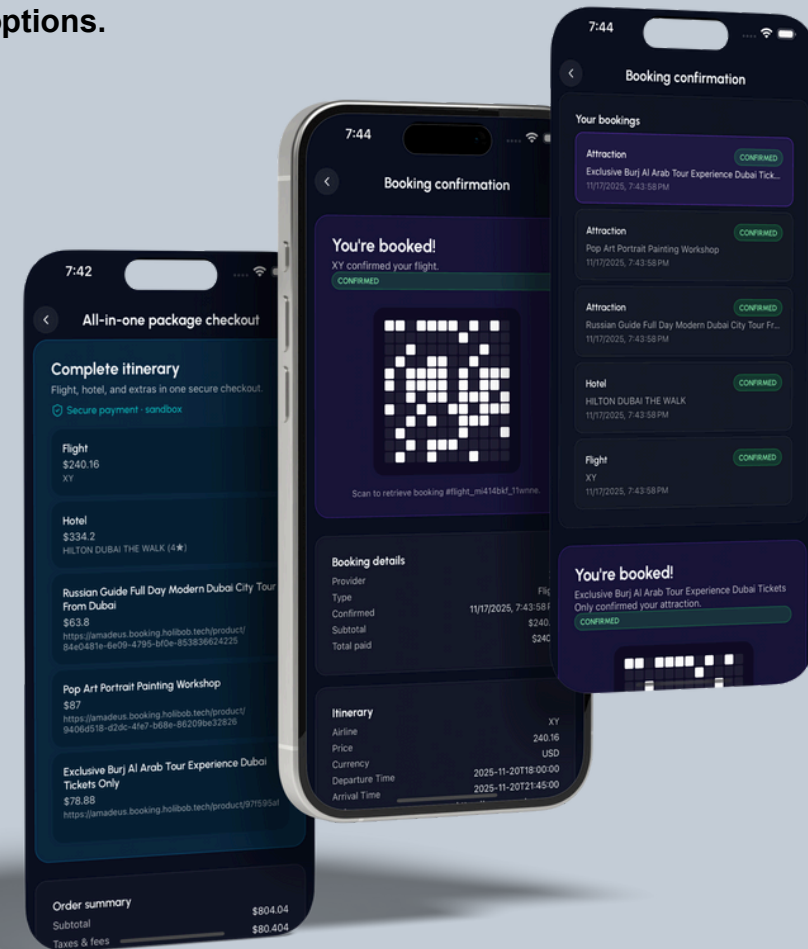
How does it work?

TWOS begins with a simple conversation. Users describe their trip in natural language, such as destination, dates, budget, and preferences, and the system automatically extracts key details.

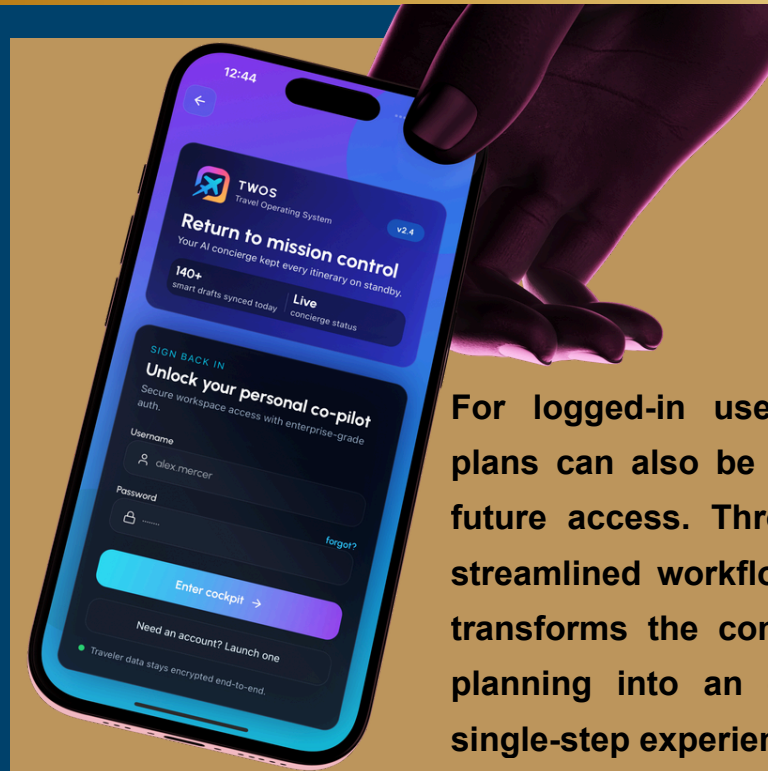


If any information is missing from the chat, the AI initiates a clarification process through a slot-filling mechanism to ensure everything needed for planning is captured accurately.

Once the requirements are complete, TWOS searches across multiple data sources, including flights, hotels, attractions, and rental car options. Its optimization engine then evaluates price, quality, user preferences, and budget constraints to generate the best-value options.

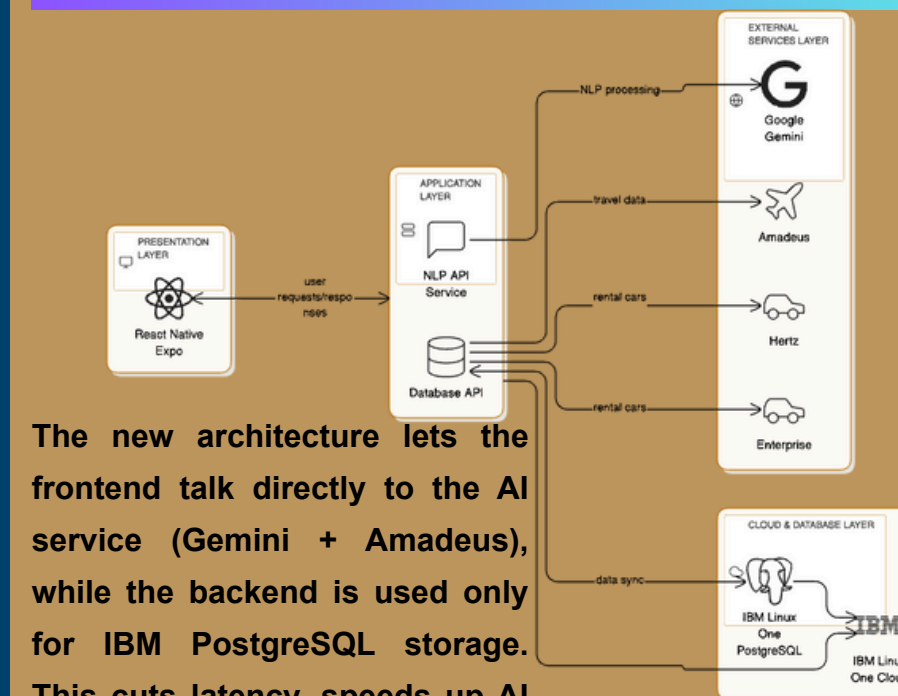


Finally, TWOS assembles a complete travel packages, presenting flights, hotels, activities, and optional car rentals in a clear, organized format.



For logged-in users, these plans can also be saved for future access. Through this streamlined workflow, TWOS transforms the complex trip planning into an effortless, single-step experience.

System Design & Architecture



The new architecture lets the frontend talk directly to the AI service (Gemini + Amadeus), while the backend is used only for IBM PostgreSQL storage. This cuts latency, speeds up AI responses, and streamlines data handling.

Summary | Conclusion

TWOS shows that planning a trip can be as simple as having a conversation. The app now builds full itineraries, checks real travel data, and completes bookings in one place. With faster architecture and in-app checkout, the experience is smooth and practical. This milestone proves the core idea works and is ready for the next stage.

Key References

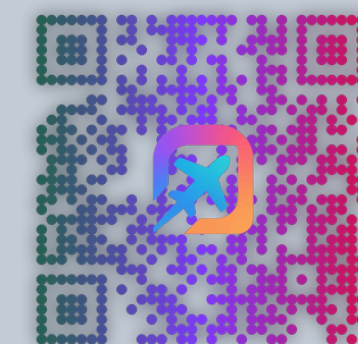
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Try it out



Acknowledgment

We thank Professor Ahmed Banafa for his guidance and support during this project. His direction helped us stay on track and refine our work.